

BARRIE-ORO, ON, Canada

WMO: 713140

Lat: **44.484N**Lon: **79.546W**Elev: **948**StdP: **14.20**Time Zone: **-5.00 (NAE)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.7	-5.3	-18.5	2.1	-11.3	-12.1	3.0	-4.4	24.4	24.6	21.6	21.4	5.0	0	0.566

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	84.9	70.6	81.9	69.1	79.3	67.5	72.8	81.3	71.1	79.1	69.6	76.6	8.9	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.1	114.7	77.4	68.4	108.2	75.7	66.9	102.6	74.3	37.1	81.7	35.6	79.2	34.2	76.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
21.2	18.8	16.5	-21.9	89.3	5.4	2.3	-25.7	91.0	-28.8	92.3	-31.9	93.6	-35.7	95.3		
			-22.2	76.2	5.3	1.5	-26.0	77.4	-29.0	78.3	-32.0	79.1	-35.8	80.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.4	17.1	18.2	27.4	41.3	53.9	62.2	67.3	65.3	58.9	47.3	35.5	24.5
	DBStd	19.84	12.97	10.81	11.35	8.51	8.40	6.53	5.33	5.30	7.14	7.95	8.99	10.25
	HDD50	4338	1021	890	703	283	52	2	0	0	9	150	438	790
	HDD65	8152	1486	1310	1164	712	358	128	37	60	209	548	885	1255
	CDD50	1925	0	0	4	21	174	369	536	475	276	67	3	0
	CDD65	266	0	0	0	0	15	45	108	71	26	1	0	0
	CDH74	2313	0	0	1	9	183	405	905	561	233	16	0	0
	CDH80	470	0	0	0	0	35	86	202	104	42	1	0	0
Wind	WSAvg	7.4	7.7	8.2	8.6	9.1	7.5	6.9	6.3	6.0	6.0	7.4	7.5	7.7
Precipitation	PrecAvg	34.3	2.6	2.2	2.3	2.6	2.9	3.3	3.1	3.1	3.2	3.0	3.0	3.0
	PrecMax	43.9	4.2	4.4	6.1	5.7	5.6	8.1	4.7	5.6	5.5	5.6	5.6	4.6
	PrecMin	26.8	1.2	0.9	0.6	0.9	1.1	0.9	1.3	1.1	1.3	1.1	1.2	1.4
	PrecStd	3.7	0.9	0.9	1.1	1.1	1.0	1.6	1.1	1.1	1.0	1.1	1.1	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	47.8	64.2	74.7	84.0	86.4	88.6	87.5	85.0	76.2	62.4	51.0
		MCWB	46.8	43.2	55.8	58.8	68.0	72.1	73.3	71.8	71.0	64.6	54.5	48.7
	2%	DB	42.5	41.1	53.4	67.0	79.5	82.5	85.2	82.7	80.1	69.9	56.5	44.9
		MCWB	40.3	38.6	47.1	53.0	64.9	69.5	70.8	69.4	68.4	60.9	51.3	42.8
	5%	DB	37.7	36.3	47.8	61.5	74.4	78.7	81.9	79.6	76.2	64.9	52.5	40.7
		MCWB	35.7	33.6	42.7	51.0	61.5	67.0	69.3	68.2	66.3	58.2	48.0	38.6
	10%	DB	34.1	33.5	42.7	56.4	70.0	75.1	79.1	76.9	72.3	60.6	48.6	37.1
		MCWB	32.4	31.4	38.4	47.1	58.6	64.3	67.4	66.4	64.2	55.6	45.1	35.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	44.4	57.5	59.9	70.2	74.1	75.6	74.5	72.9	67.6	56.0	49.2
		MCDB	47.8	46.8	62.9	68.5	80.9	83.1	85.3	82.1	81.0	73.0	60.0	50.9
	2%	WB	40.4	38.9	48.5	56.1	67.2	71.4	72.9	71.8	70.6	62.6	51.9	43.0
		MCDB	41.9	41.2	52.1	64.9	76.3	80.7	81.2	79.8	77.7	67.6	56.0	44.8
	5%	WB	35.8	34.1	43.0	52.1	64.3	69.0	71.2	70.1	68.1	59.2	48.3	39.0
		MCDB	37.5	35.9	46.9	59.5	72.3	76.1	78.9	76.8	73.8	64.0	51.8	40.6
	10%	WB	32.5	31.4	38.5	48.1	60.7	66.4	69.6	68.2	65.5	56.3	45.5	35.2
		MCDB	34.1	33.4	42.0	56.1	67.1	72.5	76.6	74.6	70.8	60.0	48.7	36.8
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.5	16.9	18.7	21.6	19.7	20.1	20.0	20.3	16.3	14.8	12.6
		MCDBR	14.7	18.5	23.0	29.2	27.4	24.1	23.2	23.5	24.9	24.2	20.7	14.7
		MCWBR	13.9	16.8	16.8	18.1	15.1	14.0	12.3	12.8	14.6	15.8	16.5	13.5
	5% WB	MCDBR	14.6	17.6	20.9	25.7	23.8	21.6	19.9	19.6	20.6	20.6	18.9	14.3
		MCWBR	14.1	16.6	16.2	18.0	14.4	13.6	11.7	12.1	13.4	15.0	15.9	13.8
Clear-Sky Solar Irradiance	taub	0.286	0.297	0.317	0.354	0.385	0.406	0.405	0.396	0.362	0.342	0.308	0.283	
	taud	2.384	2.340	2.410	2.364	2.326	2.287	2.308	2.323	2.415	2.458	2.502	2.453	
	Ebn at Noon	266	283	292	288	281	274	273	271	273	263	257	257	
	Edh at Noon	24	30	32	37	39	41	40	38	32	27	22	20	
All-Sky Solar Radiation	RadAvg	443	715	1131	1423	1711	1863	1888	1626	1283	746	445	320	
	RadStd	51	90	120	141	175	141	125	85	124	80	61	37	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page